14.01 Problem Set 4

Due at 5pm on October 20th, 2023
Late problem sets are **not** accepted.

1 True or False (20 Points)

Determine whether the following statements are True or False. Explain your answers

1. *(5 Points)* In a competitive markets, price controls are always efficient.

   F, using the supply and demand model we use we can see that any sort of price control leads to dead weight lost.

2. *(5 Points)* If individual demands are $P = 3Q_1$ and $P = 2Q_2$ then aggregate demand is $P = 5Q$, where $Q = Q_1 + Q_2$.

   False $\quad P = 3q_1$, $P = 2q_2$, $2P = 5q_2$ $\quad \frac{P}{3} + \frac{P}{2} = \frac{10P}{12}$

3. *(5 Points)* Under free entry with identical firms, the price is equal to the average cost.

   True, with no entry fee and identical firms $MC = P = AVC$

4. *(5 Points)* Free entry in a market drives down prices.

   True, because on the contrary it produces either oligopoly or monopoly, which always drive prices up.

2 Minimum Price and Demand Elasticity (30 Points)

There are two different economies, characterized by the following demand for cars

$$Q_{D1} = 2 - \frac{1}{2}p \qquad Q_{D2} = 2 - p$$

D_1 eq: $2 - \frac{P}{2} = P$, $2 - \frac{2}{3} \cdot \frac{1}{2}$, $4 = 3P$, $P^* = \frac{4}{3}$, $Q^* = \frac{1}{3}$

D_2 eq: $2 - P = P$, $P^* = 1$, $Q^* = 1$

Both economies have face the same supply of cars, given by

$$Q_S = p$$

1. *(5 Points)* Calculate the price elasticity of demand in each market.

   $PED_1 = \frac{\partial D_1}{\partial P} \cdot \frac{P}{Q_{D_1}} = -\frac{P}{4-P}$, $PED_1^* = \frac{\partial D_1}{\partial P} \cdot \frac{P^*}{Q_{D_1}^*} = -\frac{1}{2}$

   $PED_2 = \frac{\partial D_2}{\partial P} \cdot \frac{P}{Q_{D_2}} = -\frac{P}{2-P}$, $PED_2^* = \frac{\partial D_2}{\partial P} \cdot \frac{P^*}{Q_{D_2}^*} = -1$

2. *(5 Points)* Suppose these economies represent two major US cities: Boston and San Antonio. Boston is one of the most walkable cities in the US, whereas San Antonio is one of the most car dependent cities in the US. Which demand would you expect to correspond to each city?

   I would expect Boston demand to be elastic, since they are not dependent on vhicles while san antonio is to be inelastic since it almost is a necessity. Thus for Boston I expect the second demand curve $(-\frac{P}{2-P})$ while for san antonio the first one $\frac{-P}{4-P}$

3. *(5 Points)* What is the equilibrium price and quantity sold in each market?

4. *(5 Points)* Calculate the consumer and producer surplus for both markets.

   for market one: $P^* = \frac{1}{3}$, $Q^* = \frac{1}{3}$ for market 2: $P^* = 1$ $Q^* = 1$

Suppose the government decides set a minimum price of $\underline{p} = \frac{3}{2}$ to reduce car usage in order to fight climate change.

5. *(5 Points)* Find the new equilibrium price and quantity in each market. Which market had the largest change in the equilibrium quantity? How does this depend on the demand elasticity? Provide an economic intuition.

market 1:

pre price-Locle: $p^* = \frac{1}{3}$, $Q^* = \frac{1}{3}$

after: $\bar{p} = \frac{3}{2}$, $q^* = \frac{5}{4}$

Q^*_{1p} $\quad$ $Q^*_{1\bar{p}}$

$\frac{1}{6}$ $\longrightarrow$ $\frac{15}{12}$

small change due to elastic demand (high ped)

market 2:

pre price-Locle: $p^* = 1$ $q^* = 1$

after: $\bar{p} = \frac{3}{2}$, $q^* = \frac{1}{2}$

Q_{1p} $\quad$ $Q_{1\bar{p}}$

1 $\longrightarrow$ $\frac{1}{2}$

drastic change due to inelastic demand (low Ped)

this directly relies on demand elastisity (since we are changing the price and seeing how Demand reacts) the direction and correlation was explained above

6. *(5 Points)* Calculate the change in consumer surplus, producer surplus and deadweight loss from the minimum price. Which market has the largest deadweight loss in response to the minimum price? Why? Provide an economic interpretation.

firstly the first market

Consumer surplus of p^*

$$\int_0^{4/3} (4 - 2x - \frac{4}{3})\,dx = \frac{16}{9}$$

Consumer surplus of $\bar{p}$

$$\int_0^{5/4} (4 - 2x - \frac{3}{2})\,dx = \frac{50}{32}$$

the change in consumer surplus in market one:

$$\frac{50}{32} - \frac{16}{9} = \frac{-31}{144}$$

Producer surplus of p^* $\quad$ (Q_{x1} of $\bar{p}$) $\quad$ producer surplus of $\bar{p}$

$$\int_0^{4/3} (\frac{4}{3} - x)\,dx = \frac{16}{18}$$

$$\int_0^{5/4} (\frac{3}{2} - x)\,dx = \frac{35}{32}$$

the change in producer surplus in market one:

$$\frac{35}{32} - \frac{16}{18} = \frac{53}{288}$$

(tot market welfare pre $\bar{p}$) − (tot market welfare after $\bar{p}$) = Dead weight loss

$24/9 - 85/32 = 1/96$

Now the second market

Consumer surplus of p^*

$$\int_0^1 (2 - x - 1)\,dx = \frac{1}{2}$$

Consumer surplus of $\bar{p}$

$$\int_0^{1/2} (\frac{1}{2} - x)\,dx = \frac{1}{8}$$

the change in consumer surplus in market two:

$$\frac{1}{8} - \frac{4}{8} = -\frac{3}{8}$$

Producer surplus of p^*

$$\int_0^1 (1 - x)\,dx = \frac{1}{2}$$

producer surplus of $\bar{p}$

$$\int_0^{1/2} (\frac{3}{2} - x)\,dx = \frac{5}{8}$$

the change in producer surplus in market two:

$\frac{5}{8} - \frac{4}{8} = \frac{1}{8}$

(tot market welfare pre $\bar{p}$) − (tot market welfare after $\bar{p}$) = Dead weight loss

$1 - \frac{6}{8} = \frac{2}{8} = \frac{1}{4}$

the second market has a higher dead weight loss because of its higher elasticity. Because consumers respond more aprubtly to a change in price, so in more elastic markets intervations lead to less total economic welfare.

3 Market Surplus with Firm Entry (25 Points)

Consider the market for sugar. Assume the market is perfectly competitive, so all sugar producers are price takers. The demand in the US for sugar is

$$Q_D = 100 - 5p$$

The government regulates supply of sugar by not allowing foreign producers to sell in the US. Assume there are 5 producers in the US, whose long run cost function is $C(q) = \frac{1}{2}q^2$.

1. *(5 Points)* What is the individual and aggregate supply of sugar in the US?
   So individual supply = $MC = \frac{dC}{dq}(q) = q$ / aggregate = $n \cdot MC = 5p$ (n = number of firms)

2. *(5 Points)* What is the equilibrium price and quantity in this market? What is the consumer surplus and the producer surplus in equilibrium? What is the producer surplus of each individual firm?
   eq: $MC = q$ ($q = 100 - 5p$) $5p = 100 - 5p$ $p^* = 10$ $Q^* = 50$
   $q = 5p$
   demand function: $p = 20 - \frac{q}{5}$ supply function: $p = \frac{q}{5}$, $p^* = 10$, $Q^* = 50$
   $\therefore CS = \int_0^{50} 20 - \frac{q}{5} - 10\,dq = 20q - \frac{q^2}{10} - 10q \Big|_0^{50} = 500 - 250 = 250$
   $\therefore PS = \int_0^{50} 10 - \frac{q}{5}\,dq = 10q - \frac{q^2}{10} \Big|_0^{50} = 500 - 250 = 250$
   $\therefore PS_n = \frac{PS}{5} = 50$

Suppose local producers can destine their profits to put pressure on the government to avoid foreign producers to sell sugar in the US.

3. *(5 Points)* How much would the producer pay to put pressure on the government to keep the status quo and avoid entry of other competitors?
   $q^* = 10$, $p^* = 10$, $AC = \frac{C(q)}{q} = \frac{1}{2}q$
   the maximum a firm would be willing to pay is π, for each firm the $MC = q$ thus $\pi = q \cdot (p - AC) = 50$

Suppose the government passes a the Sugar Reduction Price Act, allowing the entrance of foreign producers to the market. Foreign producers have the same long run cost function $C(q) = \frac{1}{2}q^2$, and the government allows five foreign producers to enter the market.

4. *(5 Points)* Calculate the new equilibrium price and quantity in this market. What is the change in consumer surplus and producer surplus? What is the change in the producer surplus of each individual firm?
   $Q_D = 100 - 5p$ $Q_S = 10p$ $\frac{q}{10} = 20 - \frac{q}{5}$, $3q = 200$ $\therefore Q^* = \frac{200}{3}$, $p^* = \frac{200}{30}$
   $P_D = 20 - \frac{q}{5}$ $P_S = \frac{q}{10}$
   $CS = \int_0^{\frac{200}{3}} 20 - \frac{q}{5} - \frac{20}{3}\,dq = 20q - \frac{q^2}{10} - \frac{20q}{3} \Big|_0^{\frac{200}{3}} = \frac{4000}{9}$ change: $CS^{t+1} - CS^t = \frac{1750}{9}$
   $PS = \int_0^{\frac{200}{3}} \frac{200}{30} - \frac{q}{10}\,dq = \frac{20q}{30} - \frac{q^2}{20} \Big|_0^{\frac{200}{3}} = \frac{2000}{9}$ change: $PS^{t+1} - PS^t = -\frac{250}{9}$

5. *(5 Points)* True or False? Justify your answer: free entry not only transfers surplus from consumers to producers, but also increases aggregate welfare.

   false, free entry pushes the supply function downward (meaning for p we get more q) thus increases not only for welfare, but also transfers PS to CS.

4 Market Surplus with Inelastic Supply (25 Points)

Consider the rental market in Cambridge. Supply is fixed at $Q_S = 5000$ and demand is given by

$$Q_D = 10000 - p$$

1. *(2 Points)* Calculate the equilibrium price and quantity in the market, and the consumer and producer surplus.

$10000 - p = 5000 \therefore p^* = 5000, Q^* = 5000$

$$CS = \int_0^{5000} (10000 - q - 5000)\,dq = 5000q - \frac{q^2}{2}\Big|_0^{5000} = 12{,}500{,}000$$

$PS = 5000 \times 5000 = 25{,}000{,}000$

To help consumers, the government decides to subsidize rent by giving checks to consumers to help them pay rent. As a result the new demand is

$$Q_D = 12000 - p$$

2. *(5 Points)* Calculate the new equilibrium price and quantity, as well as the consumer and producer surplus.

$12{,}000 - p = 5000 \therefore p^* = 7000, Q^* = 5000$

$$CS = \int_0^{5000} (12000 - q - 7000)\,dq = 5000q - \frac{q^2}{2}\Big|_0^{5000} = 12{,}500{,}000$$

$PS = 5000 \times 7000 = 35{,}000{,}000$

3. *(5 Points)* Are consumers better off after this measure? Provide an economic intuition on why consumers are/are not better off after the subsidy (Answers that just refer to the change in consumer surplus without providing an economic interpretation will not receive full credit).

no they are not better off. Since supply is fixed (perfectly inelastic) an increase in demand will simply push p^* up to account for the new higher demand and will not give CS an increase. Furthermore the calculation done above gives more concrete proof

Suppose that instead of subsidize demand, the government decides to change zoning regulations to allow the construction of large buildings. As a result, supply increases to $Q_s = 7000$.

4. *(5 Points)* Calculate the new equilibrium price and quantity, as well as the consumer and producer surplus.

$10000 - p = 7000 \quad p^* = 3000, Q^* = 7000$

$$CS = \int_0^{7000} 10000 - q - 3000 = 7000q - \frac{q^2}{2}\Big|_0^{7000} = 24{,}500{,}000$$

$PS = 3000 \times 7000 = 21{,}000{,}000$

5. *(5 Points)* Are consumers better off after this measure? What about homeowners? Provide an economic intuition on why consumers are/are not better off after the subsidy (Answers that just refer to the change in consumer surplus without providing an economic interpretation will not receive full credit).

Consumers are better off, due to new consumers entering the market and the lowered price (higher welfare for prior participators) the CS went up.

Producers however are worse off, since all we did was shift the supply "curve" (these it's a straight line) right, they are providing $\bar{Q}$ for a lower P

6. *(3 Points)* From the government's perspective, which one of the two measures discussed in this exercise is more effective if the goal is to improve the consumers' situation? Provide an explanation.

obviously the latter option (increasing supply) would align with their goals much more. Option 2 directly addresses the consumers situation (decreasing price and increasing tenants) by shifting the supply "curve" (these it's a vertical line) to the right.

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